LESSON 13.2
SOLVING FOR UNKNOWN ANGLES

LESSON INTRODUCTION

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

LEARNING TARGETS

- I can apply angle relationships to find missing angle measures.
- I can write and solve algebraic equations to solve for unknown angles.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

ESSENTIAL QUESTIONS

- How can I find unknown angle measures by applying angle relationships?

LESSON NARRATIVE

Students build on their learning about angles from the prior lesson to apply angle relationships in solving for unknown angle measures. Students utilize their knowledge of writing and solving equations, in conjunction with their understanding of angle relationships, to write and solve equations to find unknown angles' measure.

COMPONENT SUMMARY

13.2.1 WARM-UP *(~5 MIN)*

Explain agreement or disagreement with student statements

13.2.2 GUIDED INSTRUCTION *(10 MIN)*

Use angle relationships to find missing angles

13.2.3 COLLABORATION *(15 MIN)*

Apply angle relationships to find missing angles through a collaborative activity

13.2.4 GUIDED PRACTICE *(10 MIN)*

Write algebraic equations to find missing angles

13.2.5 WRAP-UP *(~5 MIN)*

Find an unknown value and use the value to find angle measures

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Chart Paper & Markers

ADDITIONAL PREPARATION

- This lesson has several different algebraic scenarios. Work through each of them prior to facilitating with students to identify scenarios where students may have misconceptions.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not recognize the angle relationship that exists to solve the problems.
- Students may apply the wrong properties when setting up equations.
- Students may believe they need to find the value of each angle when given two vertical angles. If students have the angle measure of one of the vertical angles, the corresponding vertical angle will have the same angle measure.

THINGS TO CONSIDER

- Students have learned the angle relationships in Lesson 1. Students who did not master the content in Lesson 1 may struggle with this lesson. Consider opportunities to provide additional support for those students lacking foundational knowledge.

LITERACY AND LANGUAGE ROUTINES

Take Turns

13.2.3 Collaboration provides students with a collaborative opportunity to apply relationships to find missing angles. Students will take turns doing their column independently, and then coming back together to share their results and process with their partner.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

13.2.3 Collaboration has students working on more complex problems related to applying relationships to find missing angles. Students will decompose these complex problems into manageable parts by relating previously learned concepts to new concepts.

DIFFERENTIATE INSTRUCTION

The content of this lesson is predicated on student understanding of Lesson 1. During points of this lesson, consider working with small groups of students who need additional support based on observational and formative assessment data from Lesson 1. Strategic groupings could also provide opportunities for support. For example, pairing a student who was absent with a student who felt confident with Lesson 1. If several students struggled with the foundational knowledge from the prior day, consider offering a lesson summary from the prior day in place of 13.2.1 Warm-Up.

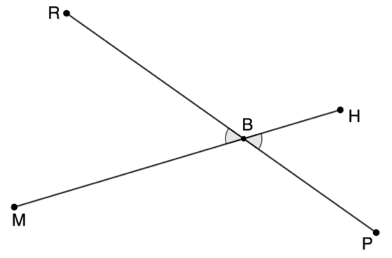
13.2.1 WARM-UP: AGREE OR DISAGREE

DIFFERENTIATE INSTRUCTION

For this activity, consider having students move to one side of the room if they agree with Jamessia, and the other side with Malik. If they don't agree with either, have them remain seated. Allow students to share out their rationale.

13.2.1 Warm-Up: Agree or Disagree

1. Line segments MH and RP intersect at point B.



a. Jamessia says $\angle MBR$ and $\angle PBH$ have the same measure. Malik disagrees and says that obviously the measure of $\angle MBR$ is larger.

Where do you stand?
- I agree with Jamessia.
- I agree with Malik.
- I disagree with both students.

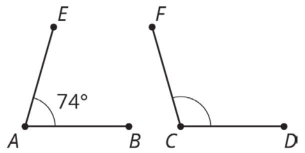
b. Explain your choice for Part A.

Answers will vary. $\angle MBR$ and $\angle PBH$ are vertical angles so they must have equal measure.

13.2.2 Guided Instruction: Applying Relationships to Find Missing Angles

Angle relationships can be used to solve mathematical problems and find missing angles.

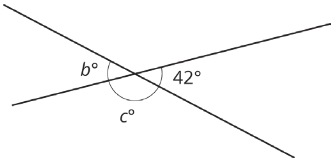
1. Angles EAB and FCD are supplementary, and the measure of $\angle EAB$ is $74°$.



Find the measure of angle FCD.

$106°$

2. Two line segments intersect, as shown.



a. Discuss with your partner the relationships between the angles marked in the diagram and how those relationships can be used to determine the missing angle measures. *Answers will vary.* $b° \cong 42°$ *since they are vertical angles.* $c° = 138°$ *since it is supplementary to* $42°$.

b. Complete the statements.

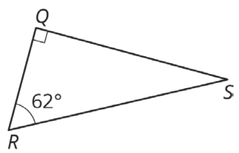
The value of b is ○ $42°$ ○ $48°$ ○ $138°$ because ○ complementary ○ supplementary ○ vertical angles have ○ the same measure. ○ a sum of $90°$. ○ a sum of $180°$.

(Answers: $42°$; vertical; the same measure.)

The value of c is ○ $42°$ ○ $48°$ ○ $138°$ because ○ complementary ○ supplementary ○ vertical angles have ○ the same measure. ○ a sum of $90°$. ○ a sum of $180°$.

(Answers: $138°$; supplementary; a sum of $180°$.)

3. In the figure, angles QRS and RSQ are complementary.



Find the measure of angle RSQ.

$28°$

13.2.2 GUIDED INSTRUCTION: APPLYING RELATIONSHIPS TO FIND MISSING ANGLES

ENRICHMENT

- What do you know about two angles that have a supplementary relationship?
- How can that be used in this scenario?

ENRICHMENT

What angle relationships do you notice from the image in question 2?

TEACHER MOVES

If students need support with breaking down the selections in each statement, allow students to share with a partner, or with the whole class, strategies they could use to attack the task of completing the statements. After hearing different strategies, allow students to complete the statements.

DIFFERENTIATE INSTRUCTION

Allow students to refer to an Anchor Chart or foldable with the definitions and examples of vertical, supplementary, and complementary angles. This is an opportunity to support students who were struggling with the vocabulary terms from Lesson 1.

ENRICHMENT

- What does it mean when two angles are complementary?
- How can you use that information to find the missing angle?

13.2.3 COLLABORATION: APPLYING RELATIONSHIPS TO FIND MISSING ANGLES

TEACHER MOVES

For this activity, consider if there are any additional instructions you might want to provide students with in completing the activity.

TEACHER MOVES

While circulating the room, look for what students are sharing out to determine if there are any major misconceptions that need to be addressed if a partner isn't catching the error.

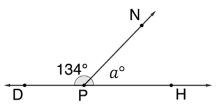
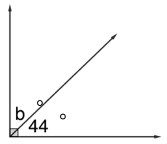
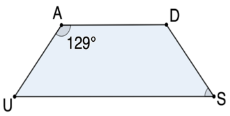
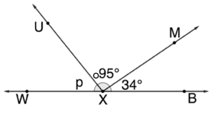
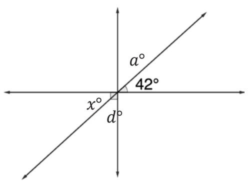
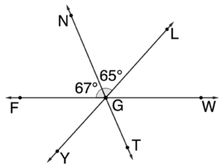
ENRICHMENT

Which strategies did you use to answer the last question when there were multiple unknown angles?

13.2.3 Collaboration: Applying Relationships to Find Missing Angles

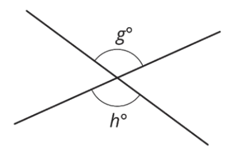
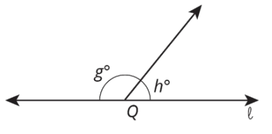
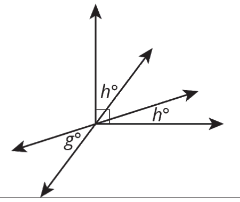
1. With your partner, determine who will be Partner A and who will be Partner B.

   a. Complete the problems in the appropriate column independently.

Partner A	Partner B
Find the value of a.  46°	Find the value of b.  46°
Angle DAU and angle USD are supplementary. Find the measure of angle USD.  51°	Find the value of p.  51°
Find the value of d.  48°	Find the measure of $\angle LGW$.  48°

   b. The problems in each row have the same result. Check your work with your partner. If there are discrepancies in your answers, work together to find and correct any errors.

2. Three figures are shown with angles measuring $g°$ and $h°$ marked in each figure.

Figure A	
Figure B	
Figure C	

Work with your partner to complete the table by matching each figure to the equation that represents the relationships between the angles in the figure. For each match, explain how you know they are a match.

Equation	Figure	Explanation
$g + h = 180$	*B*	*The angles are supplementary so they will add to 180°.*
$g = h$	*A*	*The angles are vertical so they will have the same measure.*
$2h + g = 90$	*C*	*The right angle is composed of two angles measuring $h°$ and one angle measuring $g°$ (because of vertical angles).*

13.2.3 COLLABORATION: APPLYING RELATIONSHIPS TO FIND MISSING ANGLES (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider posting these diagrams in a centralized location in the classroom on chart paper, using a projector or on an additional copy so that students can refer to them quickly as they work on the next part of the activity. Having the diagrams in a central location or projected can ease students who get frustrated and/or flustered with flipping back and forth between pages.

TEACHER MOVES

At the end of this activity, engage in a whole-group discussion, drawing specific conversation to question 2. Allow students to share out their explanations, and encourage others to agree or disagree with their explanation.

13.2.4 GUIDED PRACTICE: WRITING EQUATIONS TO FIND MISSING ANGLES

DIFFERENTIATE INSTRUCTION

For this activity, the class could be split so that half substitutes the value of x into each expression. Then, students could turn and talk with their partners to compare their answers and discuss whether they make sense given the relationship between the angles. This is an opportunity to correct potential errors as well as to recognize that the values should be equivalent.

ENRICHMENT

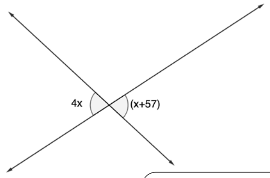
Discussion questions to consider:

- For vertical angles, do both need to be checked to find the measure of each angle?
- How could substituting the value of x into both expressions help to verify the correct angles have been determined?

13.2.4 Guided Practice: Writing Equations to Find Missing Angles

Algebraic equations can be written to express the relationship between complementary, supplementary, and vertical angles. The equations can then be solved to find the measures of unknown angles.

1. Two angles formed by intersecting lines are labeled.



a. The angles marked are
- ○ complementary angles,
- ○ supplementary angles,
- ○ vertical angles,

so they have
- ○ the same measure.
- ○ a sum of 90°.
- ○ a sum of 180°.

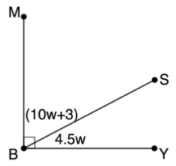
b. The equation
- ○ $4x + x + 57 = 180$
- ○ $4x + x + 57 = 90$
- ○ $4x = x + 57$

can be used to find the value of x.

c. Solve for x.
$x = 19$

d. Find the measure of each angle using the solution to Part C.
Each angle measures $76°$.

2. Angle MBY is a right angle.



a. Write an equation that can be used to find the value of w.
$10w + 3 + 4.5w = 90$

b. Solve the equation.
$w = 6$

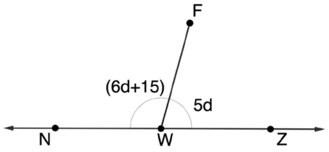
c. Use the solution to find the measure of each angle.

Angle	Measure
$\angle MBS$	$63°$
$\angle SBY$	$27°$

d. Explain whether the angle measures determined in Part C make sense based on the relationship between $\angle MBS$ and $\angle SBY$.
Answers will vary. $\angle MBS$ *and* $\angle SBY$ *are complementary angles since they form a right angle when they are adjacent. It makes sense that their angle measures add to* $90°$.

13.2.5 Wrap-Up: Ticket Out the Door

1. Two angles are labeled in the diagram of line NZ, which intersects with line segment WF.



   a. Write an equation that can be used to find the value of d.
   $6d + 15 + 5d = 180$

   b. Solve the equation.
   $d = 15$

   c. Use the solution to find the measure of each angle.

Angle	Measure
$\angle FWN$	$105°$
$\angle FWZ$	$75°$

13.2.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Proficiency on this Wrap-Up includes:

1. Fluent with classifying the types of angles.
2. Writing an equation to find a missing value, where students are knowledgeable that the two angles add to 180°.
3. Solving a multi-step equation.
4. Fluent using the value found to determine the measure of each angle.

